Wednesday 14 May 2025
14:00-15:00 BST
Duration: 1 hour
Timed exam – fixed start time

DEGREE OF MSc

Cyber Security Fundamentals M
COMPSCI5063

(Answer All Questions)

This examination paper is an open book, online assessment and is worth a total of 40 marks.

1. A company called Travelsafe operates through a website that uses URL session IDs (unique numbers assigned by a website server to a user for a specific session after the user's account has been authenticated). This website serves as an exploratory environment where visitors are provided with a search box to identify possible holiday locations. Mention and describe a potential vulnerability and the corresponding defence mechanisms that can be employed to mitigate it. [5]

2. Some companies, including Google and Amazon, use Single Sign-On (SSO) for user authentication, which enhances security. 1) Argue for the advantages and disadvantages of using SSO over traditional approaches in the given context. 2) Explain what SSO means for Confidentiality, Integrity, Authenticity, Accountability, and Non-repudiation. [7]

3. A block cipher operates on 4-bit blocks, written as hexadecimal digits (e.g., $a \oplus 9 = 3$). For one particular key K, it implements the following permutation as shown in Table 1. Using this key K, **decrypt** the following two ciphertexts according to the indicated modes of operation.

m	0	1	2	3	4	5	6	7	8	9	d	b	c	a	e	f
$E_K(m)$	1	b	5	c	7	e	2	a	4	9	f	d	0	3	6	8

Table 1: The relationship between plaintext m and its encrypted cipher text $E_K(m)$.

(a) Electronic Code Book (ECB): 187b06 [2]

(b) Cipher Block Chaining (CBC): 301be, initialization vector is 1 [4]

(c) Alice is responsible for securing sensitive company communications. To protect data transmission, she encrypts all messages using a block cipher in Electronic Code Book (ECB) mode with a secret key known only to authorized parties.

Eve, a malicious attacker, intercepts encrypted messages sent by Alice. Over time, she notices that certain ciphertext blocks repeat in different messages. Using this pattern, Eve is able to infer structural information about the original plaintext, even though she does not know the encryption key.

Explain why ECB mode encryption is not secure for protecting sensitive messages. Suggest a more secure block cipher mode and explain how it mitigates the vulnerabilities of ECB. [7]

4. **(a)** RSA (Rivest-Shamir-Adleman) is one of the first public-key cryptosystems and is widely used for secure data transmission. In RSA, the encryption key is public, and the decryption key is kept private. The public key of a given user is $e = 17$, $n = 1763$. What is the private key of this user? You must provide the steps of the process to gain full marks. [10]

(b) Assuming you can do 2^8 encryptions per second and the key size is 16 bits, how long would a brute force attack take? Give a scenario where this would be practical and another where it wouldn't. What happens if you double the key size? [5]